

Noise-Aware Qubit Mapping for Quantum Circuit Compilation

A Simulated Annealing Approach

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1. The Problem

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1.1 The Quantum Compilation Challenge

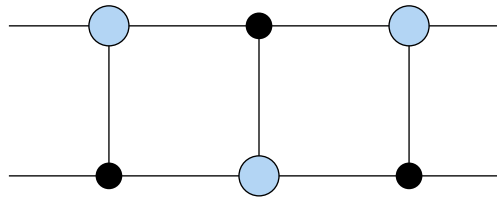
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- Two-qubit gates (e.g. CNOT) can only operate on **physically adjacent** qubits
- Quantum circuits assume **all-to-all** logical connectivity
- A **compiler** must map logical qubits to physical qubits and insert SWAP operations

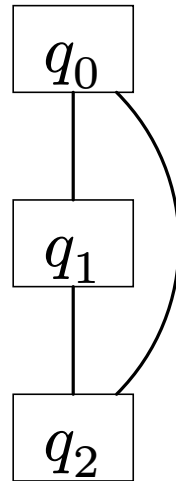
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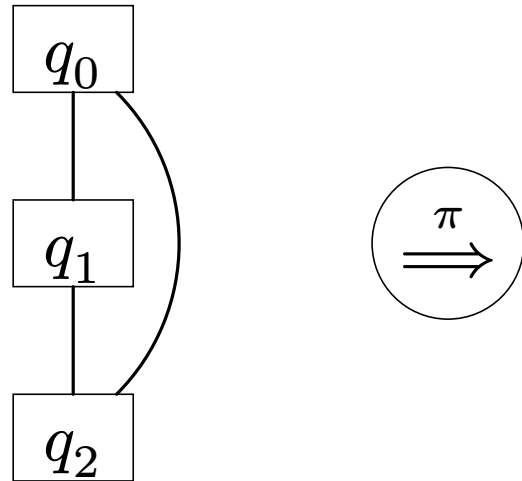
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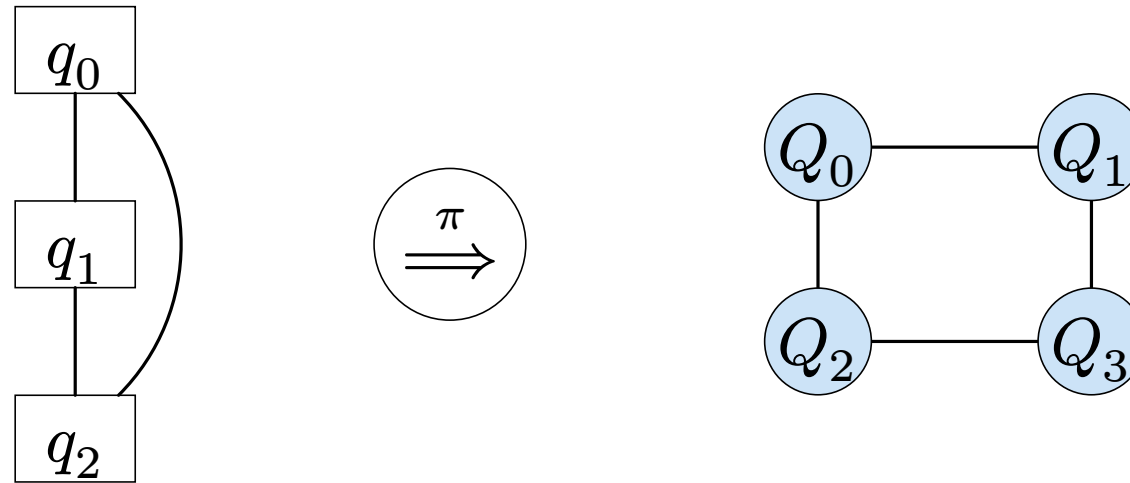


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Logical circuit (left) mapped onto physical coupling graph (right)

2. Problem Formulation

Definition 2.1.1 (Coupling Graph) A **coupling graph** $G_p = (V_p, E_p)$ represents the physical qubit topology of a quantum processor, where V_p is the set of physical qubits and E_p are the allowed two-qubit gate connections.

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Each vertex $v \in V_p$ has attributes:

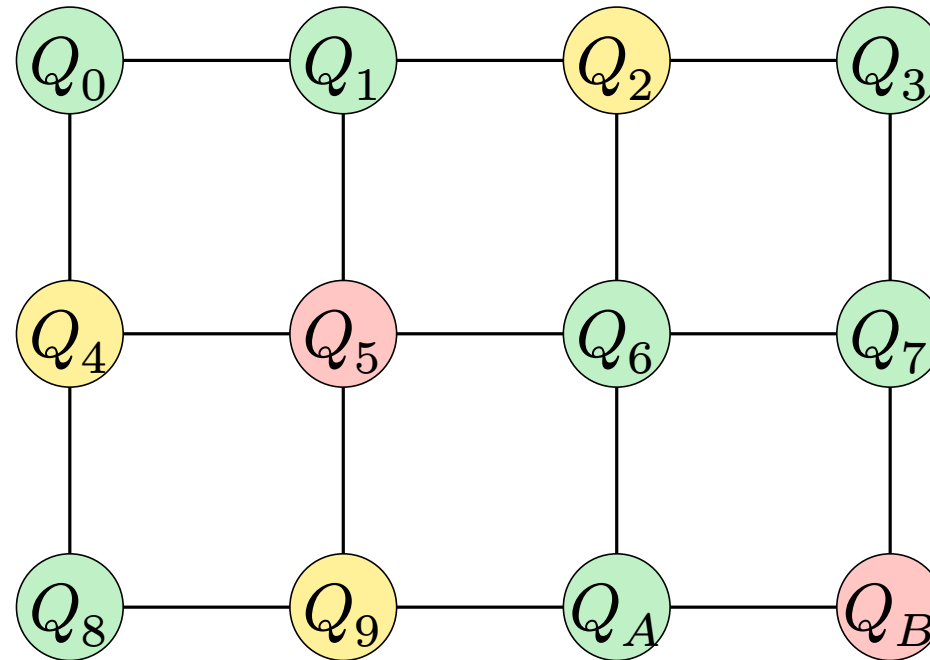
- Single-qubit error rate $\varepsilon_1(v)$ and coherence time $T_{\text{coh}}(v)$

Each edge $e \in E_p$ has a two-qubit gate error rate $\varepsilon_2(e)$.

2.1 The Coupling Graph

2.1.1 Coupling Graph Example

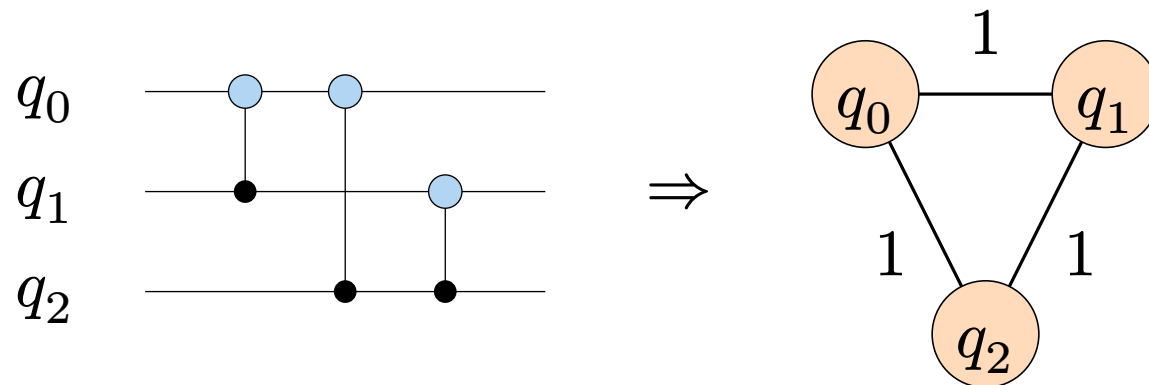
4×3 grid coupling graph (inspired by IBM devices)



Color: low error / medium / high error

Definition 2.2.1 (Interaction Graph) An **interaction graph** $G_l = (V_l, E_l, w)$ is derived from a quantum circuit, where V_l is the set of logical qubits, E_l are pairs requiring two-qubit gates, and $w : E_l \rightarrow \mathbb{N}^+$ is the gate count.

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Definition 2.2.2 (Coherence Deadline) For each logical qubit $i \in V_l$ mapped to physical qubit $\pi(i)$:

$$\text{depth}(i) \cdot t_{\text{cx}} + \text{swaps}(i, \pi) \cdot t_{\text{swap}} \leq T_{\text{coh}}(\pi(i))$$

where $\text{swaps}(i, \pi) = \sum_{j:(i,j) \in E_l} w(i, j) \cdot \max(0, d(\pi(i), \pi(j)) - 1)$.

Qubits **decohere** over time, losing quantum information. If a circuit exceeds the coherence time, the computation becomes meaningless.

3. Related Problems

3.1 Relation to Existing Problems

The Qubit Mapping Problem connects to several well-studied graph-theoretic problems:

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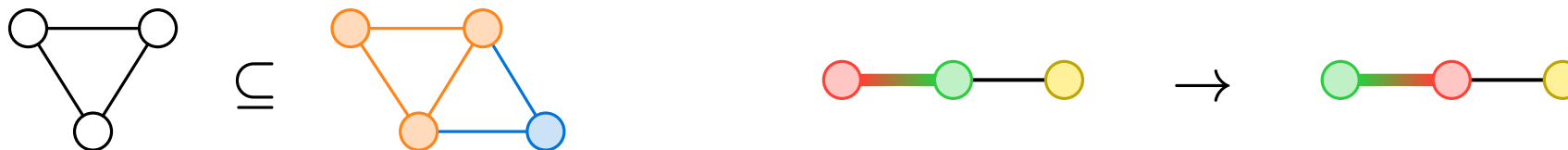
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3.1 Relation to Existing Problems

Related problems:

- **Subgraph Isomorphism** — finding G_l as a subgraph of G_p
- **Token Swapping** — minimum swaps to reach a target permutation on a graph
- **QAP** — minimising pairwise distance-weighted cost over assignments

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Our novelty: Adding **noise-aware cost** and **coherence time deadlines** transforms this from a pure graph embedding problem into a constrained multi-objective optimisation problem reflecting NISQ device limitations.

4. Objective Function

4.1 Objective Formulation

The objective minimises a weighted combination of routing cost and noise cost:

$$f(\pi) = \alpha \cdot C_{\text{route}}(\pi) + \beta \cdot C_{\text{noise}}(\pi)$$

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- **Routing cost** — total SWAP overhead:

$$C_{\text{route}}(\pi) = \sum_{(i,j) \in E_l} w(i,j) \cdot \max(0, d_{G_p}(\pi(i), \pi(j)) - 1)$$

4.1 Objective Formulation

- **Noise cost** — accumulated gate infidelity from both single and two-qubit errors:

$$C_{\text{noise}}(\pi) = \underbrace{\sum_{(i,j) \in E_l} w(i,j) \cdot \left(1 - \prod_{e \in \text{path}(\pi(i), \pi(j))} (1 - \varepsilon_2(e)) \right)}_{\text{two-qubit routing errors}} + \underbrace{\sum_{i \in V_l} \text{depth}(i) \cdot \varepsilon_1(\pi(i))}_{\text{single-qubit gate errors}}$$

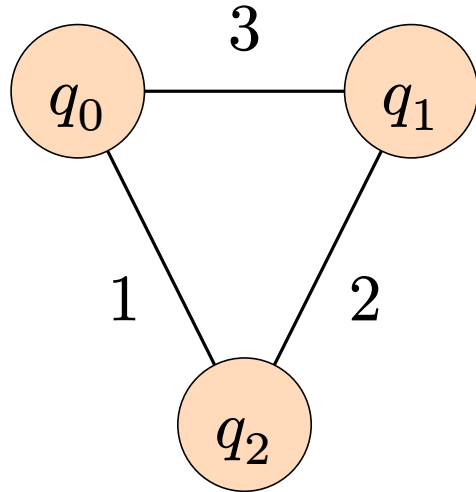
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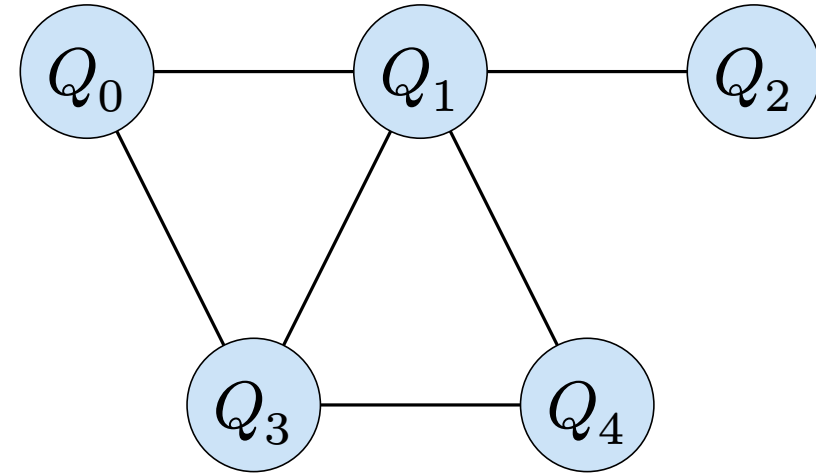
Infeasible mappings (violating coherence deadlines) receive a large additive penalty proportional to the violation magnitude.

4.2 Worked Example

Interaction Graph ($|V_l| = 3$):



Coupling Graph ($|V_p| = 5$):



4.2 Worked Example

Mapping $\pi = \{q_0 \rightarrow Q_0, q_1 \rightarrow Q_1, q_2 \rightarrow Q_3\}$:

- $C_{\text{route}} = 0$ (all pairs adjacent, $d = 1$)
- $C_{\text{noise}} = \underbrace{3\varepsilon_2(Q_0Q_1) + 1\varepsilon_2(Q_0Q_3) + 2\varepsilon_2(Q_1Q_3)}_{\varepsilon_2} + \underbrace{\sum_i \text{depth}(q_i) \cdot \varepsilon_1(\pi(q_i))}_{\varepsilon_1}$

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Mapping $\pi' = \{q_0 \rightarrow Q_0, q_1 \rightarrow Q_2, q_2 \rightarrow Q_4\}$ (all pairs $d = 2$):

- $C_{\text{route}} = 3 \times 1 + 1 \times 1 + 2 \times 1 = 6$
- C_{noise} also increases: each routing path traverses an extra edge, accumulating more ε_2 errors

5. Solution Representation

Definition 5.1.1 (Solution Representation) A solution (individual) is an **injective mapping** $\pi : V_l \rightarrow V_p$ encoded as a list where $\pi[i]$ is the physical qubit assigned to logical qubit i .

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Example: With 3 logical qubits and 5 physical qubits:

Index	$i = 0$	$i = 1$	$i = 2$
$\pi[i]$	Q_3	Q_0	Q_4

- **Constraint:** The mapping must be **injective** — no two logical qubits share the same physical qubit. This is naturally preserved by the neighbourhood operators.

5.1 Individual Encoding

- **Constraint:** The mapping must be **injective** — no two logical qubits share the same physical qubit. This is naturally preserved by the neighbourhood operators.
- **Search space size:** $\binom{n}{m} \cdot m!$ where $n = |V_p|$, $m = |V_l|$. For $n = 30$, $m = 10$: $\approx 5.6 \times 10^{13}$ possible mappings — brute force is infeasible.

6. Fitness and Algorithm

The fitness function $f(\pi)$ is evaluated as:

Step	Description
1. Precompute	Floyd-Warshall: all-pairs shortest paths $d(u, v)$ on G_p . Path error: $\text{err}(u, v) = 1 - \prod_e (1 - \varepsilon_2(e))$ for all pairs.
2. Evaluate	Sum routing cost (C_{route}) and noise cost (C_{noise} , including ε_1 and ε_2) over the mapping.
3. Penalise	Add $100 \cdot \max(0, t_{\text{total}} - T_{\text{coh}})$ for each violation.

Precomputation is $O(|V_p|^3)$ (once), each **evaluation** is $O(|E_l|)$.

Incremental evaluation after a swap: $O(\deg(i) + \deg(j))$.

6.2 Simulated Annealing

Simulated Annealing is the classical analogue of *Quantum Annealing*. The Metropolis acceptance criterion mirrors quantum tunnelling through energy barriers.

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Neighbourhood operators:

- **Swap** (60%): Exchange physical qubits of two logical qubits
- **Relocate** (40%): Move one logical qubit to an unoccupied physical qubit

Both preserve injectivity of the mapping.

6.2 Simulated Annealing

6.2.1 Cooling and Acceptance

Cooling schedule:

$$T_{k+1} = \gamma \cdot T_k, \quad \gamma = 0.9995, \quad T_0 = 50, \quad T_{\min} = 0.01$$

Acceptance criterion:

$$P(\text{accept}) = \begin{cases} 1 & \text{if } \Delta f < 0 \\ \exp(-\Delta f/T) & \text{otherwise} \end{cases}$$

Run with **5 restarts** from different random seeds, returning the best solution found.

7. Solution Decoding

7.1 From Mapping to Compiled Circuit

Given the optimal mapping π^* , the solution is decoded back to a real-world quantum circuit:

1. **Assign** each logical qubit q_i to physical qubit $Q_{\pi^*(i)}$
2. **Route** each two-qubit gate: if $d(\pi^*(i), \pi^*(j)) > 1$, insert SWAP gates along the shortest path to bring the qubits adjacent
3. **Verify** that the total circuit depth respects coherence time limits

7.1 From Mapping to Compiled Circuit

Quality metrics of the compiled circuit:

Metric	Meaning
Total SWAPs	Number of inserted SWAP operations (each = 3 CNOTs)
Circuit fidelity	$\prod (1 - \varepsilon)$ over all gates — probability of correct execution
Feasibility	Whether all qubits finish before their coherence deadline

Thank you!

Any questions?